Zero- Flow Engine

Beginner Setup Guide — Laptop / CPU Mode (no NVIDIA GPU)

Windows 11 · Lenovo IdeaPad Slim 5 (i7-13620H, 16 GB RAM, Intel UHD) · github.com/zerocold7/local-voice-flow

Zero- Flow is a private, on-device voice typing tool. You hold a key, speak, and your words are typed into whatever app you are using — in English or Arabic. Nothing is uploaded; it all runs on your laptop. This guide takes you from a fresh machine to a working setup, one step at a time. Just follow the sections in order.

What you need before you start

- Your laptop, signed in to Windows 11.
- An internet connection — you will download about 6–8 GB in total (one time only).
- About 30–45 minutes — most of that is just waiting for downloads.
- Roughly 10 GB of free disk space (you have plenty).
- Your built-in microphone — no extra hardware needed.

> **TIP** Your laptop has no NVIDIA graphics card, and that is completely fine. This guide sets up **CPU mode**. It works exactly the same; transcription just takes about 1–4 seconds after you release the key instead of being instant.

Step 1 — Open PowerShell

PowerShell is the window where you type the setup commands.

1. Click the **Start** button (Windows logo, bottom-left).
2. Type the word **PowerShell**.
3. Click **Windows PowerShell** in the results. A dark window opens.

You will type (or paste) one command at a time and press **Enter** after each. To paste a command you copied, **right-click** inside the window.

Step 2 — Install the three required programs

You need three free programs: **Python** (runs the engine), **Git** (downloads the project), and **Ollama** (the local AI). Paste these three lines into PowerShell, pressing Enter after each:

```
winget install Python.Python.3.12
winget install Git.Git
winget install Ollama.Ollama
```

When they finish, **close PowerShell and open it again** (Step 1) so it notices the new programs. Then check they all installed by running:

```
python --version
git --version
ollama --version
```

Each line should print a version number (for example `Python 3.12.x`). If so, you are good.

TIP If you see **'winget is not recognized'**: open the **Microsoft Store**, search for **App Installer**, and click **Update**. Then close and reopen PowerShell and try again.

TIP If typing **python** opens the **Microsoft Store** instead of printing a version: go to **Settings > Apps > Advanced app settings > App execution aliases** and turn **OFF** the two entries named 'python'. (Or simply type `py` instead of `python` everywhere in this guide.)

Step 3 — Turn on the microphone (do not skip)

Windows hides the microphone from desktop apps by default. If you skip this, the engine will record silence and type nothing.

1. Open **Settings** (Start menu > the gear icon).
2. Go to **Privacy & security > Microphone**.
3. Turn **ON** ‘Microphone access’.
4. Turn **ON** ‘Let desktop apps access your microphone’.

Step 4 — Download the project

Back in PowerShell, run these lines. They put the project in a folder called **local-voice-flow** inside your Documents:

```
cd $HOME\Documents
git clone https://github.com/zerocold7/local-voice-flow.git
cd local-voice-flow
```

Step 5 — Two small edits for a laptop without NVIDIA

Edit A — remove the GPU-only lines. Open the requirements file:

```
notepad requirements.txt
```

Scroll to the bottom and **delete the last two lines** (`nvidia-cublas-cu12` and `nvidia-cudnn-cu12`). Save with **Ctrl+S**, then close Notepad. These are only for NVIDIA cards and would waste ~1.2 GB.

Edit B — choose laptop-friendly AI models. Create your settings file and open it:

```
Copy-Item .env.example .env
notepad .env
```

Change these two lines so they read exactly like this, then save (Ctrl+S) and close:

```
WHISPER_MODEL_NAME="small"
FALLBACK_LLM="qwen2.5:3b"
```

TIP Why: the defaults (`large-v3` and `gemma2:27b`) are built for a big GPU and lots of memory. `small` + `qwen2.5:3b` fit comfortably in your 16 GB laptop.

Step 6 — Install the engine's Python parts

These commands create a private folder named **venv** and download the needed libraries into it. Run them one by one (the last one takes a few minutes — wait until the prompt returns):

```
.\venv\Scripts\python.exe -m pip install --upgrade pip
.\venv\Scripts\python.exe -m pip install -r requirements.txt
```

(The first command below creates the **venv** folder — run it before the two above:)

```
python -m venv venv
```

TIP Run the three commands in this order: first `python -m venv venv`, then the two `pip install` lines.

Step 7 — Download the local AI model

Ollama runs quietly in the background after install (look for its icon near the clock). Download a small bilingual model:

```
ollama pull qwen2.5:3b
```

This is about 2 GB, one time only. Lighter option: `gemma2:2b`. Higher quality but slower: `qwen2.5:7b` (if you choose that, also put its name in `FALLBACK_LLM` in your `.env`).

Step 8 — Start it for the first time

1. Open **File Explorer** and go to **Documents > local-voice-flow**.
2. Double-click **Launch_Flow.bat**.

IMPORTANT If Windows shows a blue **'Windows protected your PC'** box, click **More info**, then **Run anyway**. This is just Windows being cautious about a new file — it is your own project file.

A console window opens. The **first** launch downloads the Whisper 'small' model (~500 MB) — let it finish. When you see a line like this, it is ready:

```
Whisper model active on CPU (int8)
```

Leave that window open (you can minimize it). **Closing the window stops the engine.**

Step 9 — Test that it works

1. Open **Notepad** and click inside it so the cursor is blinking.
2. Hold **F5**, say 'hello, this is a test', then release F5.
3. After a second or two, your words appear in Notepad.
4. Try **F7** and speak Arabic the same way.

If nothing appears, see the Troubleshooting table near the end.

Step 10 — Hotkey reference

Hold the key, speak, release. The key forces the language, so it is never misheard.

Key	What it does
F5	Dictate English — raw (exactly as spoken)
F6	Dictate English — polished (AI cleans grammar/fillers)
F7	Dictate Arabic — raw
F8	Dictate Arabic — polished
F9	Translate English -> Arabic
F10	Translate Arabic -> English
Shift + F1	Tidy up the learned-vocabulary file
Shift + F2	Clear the debug log and dictation history
Shift + F3	Rewrite and fix the current line
Esc	Cancel the recording in progress

Step 11 — Everyday use: starting and stopping

- **Start:** double-click `Launch_Flow.bat` and minimize the window.
- **Stop:** close that console window (click the X).
- **Find the tray icon:** click the small **^** arrow next to the clock; the Zero- Flow icon lives there.
- **Ollama** starts automatically with Windows, so the AI features just work.

> **TIP** Optional — run it invisibly: double-click `Launch_Silent.vbs` to start with no window. To stop it then, open **Task Manager** (Ctrl+Shift+Esc), find **python**, and click **End task**. Stick with the visible `.bat` until you are comfortable.

Troubleshooting

If this happens...	Do this
Nothing is typed when I speak	Check microphone permissions (Step 3). Make sure you held the key the whole time you spoke. Look at the console window for a red error message.
'python is not recognized'	Close and reopen PowerShell. If it still fails, reinstall Python and tick 'Add to PATH', or type `py` instead of `python`.
'winget is not recognized'	Update **App Installer** from the Microsoft Store, then reopen PowerShell.
It types in the wrong language	Use the matching key: F5/F6 for English, F7/F8 for Arabic. The key decides the language.
Hotkeys do nothing in one specific app	Right-click `Launch_Flow.bat` and choose **Run as administrator**.
Polish / Translate does nothing	Make sure Ollama is running: in PowerShell type `ollama list` and confirm your model is listed (Step 7).
Everything feels too slow	In `.env` set `WHISPER_MODEL_NAME="base"`, save, and restart the engine.

If this happens...	Do this
SmartScreen blocks the launcher	Click **More info**, then **Run anyway**.

Making it faster or more accurate

After changing any of these in `.env`, save the file and **restart the engine** (close the console and run the launcher again).

Speech model (`WHISPER_MODEL_NAME`):

Value	Trade-off
tiny / base	Fastest, lowest accuracy. Good if 'small' feels slow.
small	Recommended balance for your laptop (this guide's default).
medium	Better Arabic accuracy, noticeably slower on CPU.

AI model (`FALLBACK_LLM`, pulled with `ollama pull`):

Value	Trade-off
gemma2:2b	Lightest, fastest.
qwen2.5:3b	Recommended balance, good Arabic + English.
qwen2.5:7b	Best quality, slower on CPU.

Privacy: everything in this guide runs locally on your laptop — your voice and text are never uploaded. For deeper customization (other languages, other AI providers, prompts), see **CONFIGURATION.md** inside the project folder.